

Predictive Information Structure in Equity Markets: An Interpretable Machine Learning Perspective

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Abstract

While much of the financial prediction literature has focused on improving forecasting performance through increasingly sophisticated machine learning models, comparatively less attention has been devoted to understanding how predictive information is organized across interpretable feature families. This study investigates the organization of predictive information in Tehran Stock Exchange equities using an interpretable financial machine learning framework. The proposed framework organizes variables constructed from daily price and volume data covering the period from March 26, 2022 to November 21, 2025 into five feature families to examine predictive relevance across the feature space.

The results reveal a clear concentration of predictive attribution across feature families. Daily Price Structure receives the largest aggregate attribution share, followed by momentum and volume flow and liquidity families. Cross method explainability analysis shows identical family rankings under permutation importance and SHapley Additive exPlanations (SHAP) attribution. Out of sample ablation analysis further shows that, among Daily Price Structure dominated equities, the dominant configuration retains standalone predictive performance, remaining comparable to, and on average slightly stronger than, the full feature configuration, while exclusion of the dominant family substantially reduces predictive performance.

The study contributes to interpretable financial machine learning by introducing a feature family perspective for jointly examining model attribution and subsequent predictive performance. Within the examined market, feature space, and prediction setting, the findings indicate that predictive relevance is concentrated rather than uniformly distributed across financial feature families.

Keywords: Explainable Artificial Intelligence, Financial Machine Learning, Feature Families, Predictive Information, Interpretable Machine Learning, Stock Markets

1 Introduction

Machine learning methods have become increasingly influential in financial forecasting due to their ability to capture nonlinear relationships, high dimensional interactions, and complex market dynamics that are difficult to model using traditional econometric approaches (Gu, Kelly, & Xiu, 2020; Hoepner, McMillan, Vivian, & Simen, 2021; Weber, Carl, & Hinz, 2024; Yeo et al., 2025). In equity markets, these methods are widely used for return prediction, directional forecasting, portfolio construction, volatility estimation, and risk management (Fischer & Krauss, 2018; Gu et al., 2020; Sezer, Gudelek, & Ozbayoglu, 2020; Sirignano & Cont, 2019). As machine learning models have become increasingly complex, concerns regarding transparency, interpretability, robustness, and model accountability have received growing attention in financial research (Černevičienė & Kabašinskas, 2024; Hoepner et al., 2021; Weber et al., 2024; Yeo et al., 2025).

Many machine learning models operate as black box systems whose underlying decision processes remain difficult to interpret (Lundberg & Lee, 2017; Ribeiro, Singh, & Guestrin, 2016). Although these models often achieve strong predictive performance, predictive performance alone provides limited understanding of how financial information contributes to model decisions (Hoepner et al., 2021; Weber et al., 2024). Consequently, explainable artificial intelligence (XAI) has become an important component of financial machine learning by providing tools for interpreting model behavior, evaluating feature relevance, and improving confidence in model based decision making (Černevičienė & Kabašinskas, 2024; Weber et al., 2024; Yeo et al., 2025).

Despite these advances, an important question remains insufficiently explored. Existing studies have primarily examined explainability at the level of individual features, while comparatively less attention has been devoted to understanding how predictive information is organized across broader and more interpretable feature families (Černevičienė & Kabašinskas, 2024; Weber et al., 2024; Yeo et al., 2025). This limitation is particularly relevant in financial prediction because many financial indicators capture related aspects of market behavior. As a result, interpretation based solely on individual feature rankings may provide only a partial representation of predictive structure when predictive information is distributed across correlated variables (Aas, Jullum, & Løland, 2021; Owen & Prieur, 2017; Strobl, Boulesteix, Zeileis, & Hothorn, 2007).

Motivated by this limitation, the present study investigates how predictive information is distributed across interpretable feature families within the final modeling universe of Tehran Stock Exchange (TSE) equities. More specifically, the study examines whether the distribution of predictive attribution is consistent across complementary explainability methods and whether the family identified as dominant can, on its own, preserve the predictive performance of the full feature set on unseen data.

To address these questions, the study develops a unified interpretable machine learning framework for examining how predictive information is organized across feature families and whether attributional dominance translates into standalone predictive value. The main contributions of the study are as follows.

1. The study introduces a feature family aggregation procedure for evaluating how predictive information is distributed across feature families. Features are organized into families associated with momentum, volatility, volume flow and liquidity, trend structure, and Daily Price Structure, allowing predictive structure to be examined across broader dimensions of market information.
2. The study develops a triangulated explainability framework that integrates SHAP attribution, train only permutation importance, and cross method rank consistency analysis (Lundberg & Lee, 2017; Weber et al., 2024). The combination of complementary explainability procedures reduces reliance on a single attribution method and provides a more comprehensive assessment of feature family importance.
3. The study introduces an out of sample dominant versus non dominant ablation procedure for evaluating whether dominant feature families retain standalone predictive information on unseen data. An exploratory clustering analysis additionally examines whether variation in feature family attribution profiles exhibits systematic correspondence with industry classifications.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and identifies the research gap. Section 3 describes the dataset, feature construction, and feature family aggregation. Section 4 presents the empirical methodology. Section 5 reports the empirical results. Section 6 discusses the findings. Section 7 outlines the study limitations and future research directions. Finally, Section 8 concludes the paper.

2 Related Work and Research Gap

2.1 Financial Machine Learning Beyond Predictive Accuracy

Forecasting financial markets remains one of the central challenges in quantitative finance due to the nonlinear, dynamic, and evolving nature of financial systems. Advances in computational power and the availability of large financial datasets have led to the widespread adoption of machine learning techniques for financial prediction, complementing traditional statistical and econometric approaches (Bollerslev, 1986; Box, Jenkins, Reinsel, & Ljung, 2015; Fama & French, 1993; Fischer & Krauss, 2018; Gu et al., 2020; Hoepner et al., 2021; Sezer et al., 2020; Sirignano & Cont, 2019). Compared with conventional econometric models, machine learning methods provide greater flexibility for modeling nonlinear relationships and complex interactions among financial variables, supporting a wide range of financial prediction tasks including stock prediction, volatility forecasting, portfolio management, fraud detection, algorithmic trading, and market microstructure analysis (Fischer & Krauss, 2018; Gu et al., 2020; Sezer et al., 2020; Sirignano & Cont, 2019; Weber et al., 2024; Yeo et al., 2025).

Despite these advances, recent studies increasingly argue that progress in financial machine learning should not be evaluated solely through improvements in predictive

accuracy (Černevičienė & Kabašinskas, 2024; Hoepner et al., 2021; Weber et al., 2024; Yeo et al., 2025). Because financial markets evolve over time, strong predictive performance alone provides limited understanding of how prediction models utilize financial information or whether the learned relationships remain meaningful under changing market conditions (Hoepner et al., 2021; Weber et al., 2024; Yeo et al., 2025). Consequently, recent research has placed growing emphasis on robustness, transparency, and the interpretation of predictive mechanisms in addition to forecasting performance. This shift has established XAI and interpretable machine learning as important complementary directions for understanding financial prediction models beyond predictive accuracy (Černevičienė & Kabašinskas, 2024; Weber et al., 2024; Yeo et al., 2025).

2.2 Explainability in Financial Machine Learning

XAI has become an integral component of financial machine learning by providing tools for interpreting model behavior, evaluating feature relevance, and improving confidence in model based decision making (Černevičienė & Kabašinskas, 2024; Weber et al., 2024; Yeo et al., 2025). The increasing availability of post hoc explanation methods has substantially improved the interpretation of complex prediction systems and supported feature evaluation, robustness assessment, and model validation across a wide range of financial applications (Černevičienė & Kabašinskas, 2024; Weber et al., 2024; Yeo et al., 2025).

Among post hoc explainability methods, SHAP, permutation importance, and LIME are widely adopted because they can be applied across diverse machine learning algorithms (Lundberg & Lee, 2017; Ribeiro et al., 2016; Weber et al., 2024). Despite these advances, much of the existing literature continues to focus on attribution at the level of individual features.

2.3 From Individual Features to Feature Families

Financial indicators rarely represent completely independent sources of predictive information. Instead, many variables capture related aspects of the same underlying market conditions and therefore exhibit substantial overlap in the information they convey (Aas et al., 2021; Owen & Prieur, 2017; Strobl et al., 2007). For example, momentum variables often reflect similar persistence effects, while liquidity measures, trading activity, and daily price characteristics collectively describe related dimensions of market information. Under these circumstances, interpretation based solely on individual feature rankings may provide only a fragmented representation of predictive relevance within financial datasets (Aas et al., 2021; Owen & Prieur, 2017; Strobl et al., 2007; Weber et al., 2024).

This challenge becomes more pronounced in financial machine learning because prediction models frequently incorporate large collections of correlated variables. Attribution methods may distribute predictive relevance across multiple related features even when those features represent similar dimensions of market information, leading to rankings that vary across models, datasets, or explainability methods (Aas et al., 2021; Owen & Prieur, 2017; Strobl et al., 2007). Consequently, interpretation at the

level of individual features may not adequately capture the broader organization of predictive information within financial feature spaces.

These observations motivate examining predictive information from a broader perspective. Rather than interpreting variables individually, related features can be considered collectively according to the dimensions of market information they represent, providing a more coherent representation of predictive structure (Weber et al., 2024; Yeo et al., 2025). This perspective is particularly relevant in financial applications because many variables are designed to quantify related dimensions of market information. Examining such variables collectively therefore provides a basis for investigating the distribution of predictive information beyond isolated feature rankings.

2.4 Research Gap

Existing research has substantially advanced financial machine learning by improving predictive models, expanding explainability methods, and enhancing the interpretation of financial prediction systems. Nevertheless, important limitations remain. Current studies continue to emphasize predictive performance and attribution at the level of individual features, while comparatively limited attention has been devoted to understanding how predictive information is organized across related groups of financial variables. Under correlated financial feature spaces, isolated attribution scores may provide an incomplete or unstable representation of predictive relevance when multiple variables describe related dimensions of market information (Aas et al., 2021; Černevičienė & Kabašinskas, 2024; Owen & Prieur, 2017; Strobl et al., 2007; Weber et al., 2024).

Relatively limited evidence is also available on whether feature family attribution patterns remain consistent across complementary explainability methods. Moreover, identifying a family as dominant in terms of predictive attribution does not necessarily establish that the same family can preserve the predictive performance of the full feature set when evaluated independently on unseen data. This distinction between attributional dominance and standalone predictive value remains comparatively under-explored. The TSE additionally provides a setting in which differences in liquidity conditions, retail participation, information diffusion, and trading behavior may influence how predictive information is distributed across market derived features (Bekaert & Harvey, 2003; Harvey, 1995). Taken together, these limitations indicate that how predictive information is distributed and organized across financial feature families remains insufficiently understood.

3 Data and Feature Family Framework

3.1 Dataset Construction

The empirical analysis is conducted using daily data from the TSE. The dataset consists of adjusted daily price and volume data, including adjusted open, high, low, close, and final prices together with trading volume, covering the period from March 26, 2022 to November 21, 2025. The study focuses exclusively on common equities

satisfying the data quality and predictive feasibility criteria defined by the two stage eligibility screening procedure described in Section 4.1.

The initial universe included a broad cross section of TSE equities spanning multiple industries and liquidity profiles. Because many securities in emerging markets exhibit substantial illiquidity, trading interruptions, low turnover, or structurally unstable price behavior, additional feasibility filtering procedures were required before inclusion in the empirical analysis. All preprocessing and feature generation procedures preserved chronological ordering, and adjusted price series were used throughout the analysis to maintain consistency around corporate actions.

3.2 Feature Engineering

The feature engineering process was designed to construct 37 predictive variables from daily adjusted price and volume data while preserving chronological ordering. The constructed feature space combines classical technical indicators with price derived, volume derived, rolling statistical, and candlestick derived variables. All features were generated through deterministic rule based procedures using only information available up to each observation date.

The engineered features include lagged adjusted prices, multi horizon percentage price changes, rolling means, rolling standard deviations, standardized price deviations, daily price range measures, and volatility related transformations. In addition, the feature set includes widely used technical indicators such as Rate of Change (ROC), Relative Strength Index (RSI), Average True Range (ATR), moving averages, Moving Average Convergence Divergence (MACD), On Balance Volume (OBV), Money Flow Index (MFI), and Chaikin Money Flow (CMF) (Achelis, 2001; Murphy, 1999).

The feature space also includes candlestick and daily price derived variables constructed directly from adjusted open, high, low, close, and final prices. These variables characterize candlestick body magnitude, upper and lower shadow lengths, closing location within the daily trading range, and the difference between closing and final adjusted prices. The final price is a session level reference price distinct from the price of the last transaction. Candlestick representations have been used to describe short term stock price patterns and to support stock market timing decisions, motivating their inclusion in the engineered feature set (Lee & Jo, 1999).

All rolling transformations and technical indicators were computed in a time safe manner using only current and historical observations. Ratio based variables were computed using a safe division procedure to avoid undefined numerical operations. Detailed mathematical definitions of all engineered variables are provided in Appendix A.

3.3 Feature Family Definition and Aggregation

To investigate the organization of predictive information beyond individual feature attribution, all engineered features were organized into five mutually exclusive feature families: momentum, volatility, Volume Flow and Liquidity, trend, and Daily Price Structure. This organization enables predictive information to be evaluated across broader and more interpretable groups of financial features rather than isolated individual variables.

The proposed feature family taxonomy represents an operational organization of the engineered feature space developed for the objectives of the present study. The selected families are consistent with major categories of market analysis discussed in the technical analysis and financial forecasting literature, including momentum, volatility, trend, volume based analysis, and candlestick based price behavior (Achelis, 2001; Lee & Jo, 1999; Murphy, 1999). The taxonomy should therefore be interpreted as a study specific taxonomy of OHLCV derived features rather than as a universal classification of financial indicators.

The five families were selected to organize the main types of information represented in the constructed feature space. Momentum features describe directional persistence and recent price movement (Achelis, 2001; Murphy, 1999). Volatility features describe price dispersion and short term variability (Achelis, 2001; Murphy, 1999). Volume Flow and Liquidity features describe trading participation and money flow dynamics using volume derived variables (Achelis, 2001). Trend features describe lower frequency directional structure using smoothing and trend following indicators (Achelis, 2001; Murphy, 1999). Daily Price Structure features describe the configuration of prices within the daily trading interval using candlestick and related price derived variables (Lee & Jo, 1999).

Table 1 summarizes the five feature families together with representative engineered features used throughout the empirical framework.

Each engineered feature was assigned to a single feature family according to its dominant financial interpretation. Although some features may be statistically related to more than one dimension, mutually exclusive assignment was adopted to preserve interpretability and enable consistent aggregation of explainability measures. Feature importance values obtained from the explainability methods were subsequently aggregated at the family level using normalized importance scores. This aggregation procedure enables comparison of predictive relevance across feature families while reducing reliance on isolated importance rankings for correlated individual variables.

4 Methodology

4.1 Data Quality and Feasibility Screening

The empirical framework employs a two stage eligibility screening procedure to identify equities with sufficient data quality, trading feasibility, and predictive information for subsequent modeling and interpretability analysis. The purpose of the screening procedure is not to maximize forecasting performance, but to restrict the analysis to equities for which the available data and feature space provide an adequate basis for meaningful interpretation of predictive information.

The first screening stage evaluates raw data integrity and trading feasibility using adjusted daily price and volume data for each equity. Diagnostic measures include trading calendar coverage, missing value frequency, zero volume occurrence, duplicate timestamps, locked range behavior, and feature generation feasibility. A preliminary predictive feasibility assessment is additionally conducted using a lightweight logistic regression model evaluated through time aware cross validation. Only equities satisfying the predefined data quality, trading feasibility, and preliminary predictive

Table 1: Feature Family Taxonomy Used in the Empirical Framework

Feature Family	Representative Feature Types	Financial Interpretation
Momentum	Lagged adjusted prices, multi horizon percentage price changes, price momentum, Rate of Change (ROC), and Relative Strength Index (RSI)	Recent directional movement and persistence of price changes
Volatility	Daily price range, Average True Range (ATR), rolling standard deviation, standardized price deviation, and relative range expansion	Short horizon dispersion, variability, and unusual price range behavior
Volume Flow & Liquidity	Volume changes, volume to rolling average ratios, On Balance Volume (OBV), Money Flow Index (MFI), Chaikin Money Flow (CMF), and volume weighted price	Trading activity, market participation, and price-volume flow dynamics
Trend	Rolling and simple moving averages, Moving Average Convergence Divergence (MACD), MACD signal, and recent directional consistency	Smoothed directional structure and lower frequency price behavior
Daily Price Structure	Candlestick body magnitude, closing to final price difference, upper and lower shadow lengths, and closing location within the daily range	Within day price configuration and relative positioning of open, high, low, close, and final prices

Notes: The feature space contains 37 engineered variables assigned to five mutually exclusive feature families according to their dominant financial interpretation. The families contain 13 Momentum, 5 Volatility, 9 Volume Flow and Liquidity, 5 Trend, and 5 Daily Price Structure variables. Detailed feature definitions and assignments are provided in Appendices A and B. Abbreviations: ROC = Rate of Change; RSI = Relative Strength Index; ATR = Average True Range; OBV = On Balance Volume; MFI = Money Flow Index; CMF = Chaikin Money Flow; MACD = Moving Average Convergence Divergence.

feasibility criteria proceed to the second screening stage. The corresponding decision thresholds are reported in Appendix A.

Table 2 summarizes the screening criteria applied during the initial quality and feasibility screening stage.

The second screening stage evaluates the engineered feature space described in Section 3.2. Equities failing the predefined feature quality, stability, or predictive feasibility criteria are excluded from the final modeling universe. The corresponding decision thresholds are reported in Appendix A. This stage is completed before the subsequent explainability and ablation analyses, and neither explainability measures nor outcomes from the internal test set are used to determine feature family dominance.

The resulting final modeling universe therefore consists of equities satisfying both stages of the eligibility screening procedure. The subsequent feature family analysis should accordingly be interpreted as conditional on this screened universe rather than as an unconditional characterization of all TSE equities. The overall empirical workflow is summarized in Figure 1.

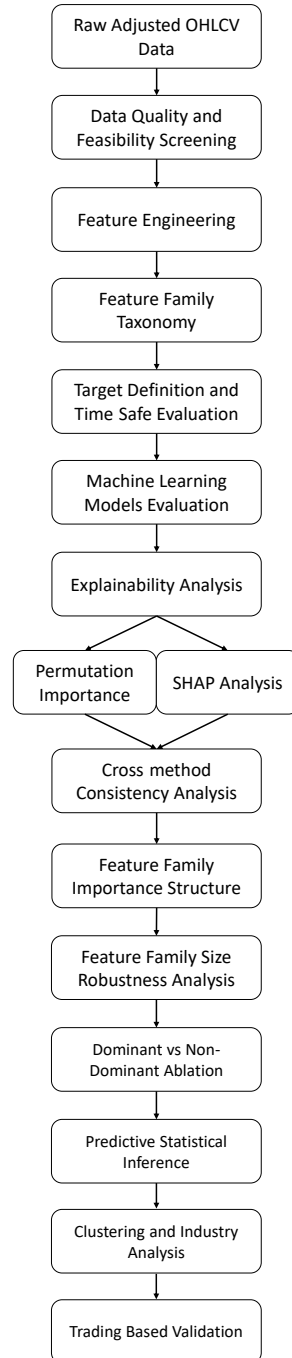


Fig. 1: Overall Empirical Framework of the Proposed Study

Table 2: Multi Stage Data Quality and Feasibility Screening Procedure

Screening Component	Description	Purpose
Historical coverage	Sufficient historical observations for model estimation	Ensure temporal continuity
Missing values	Frequency of unavailable observations	Ensure sufficient data completeness
Zero volume days	Frequency of non trading observations	Identify illiquid equities
Timestamp consistency	Detection of duplicate trading records	Preserve data integrity
Locked price behavior	Detection of prolonged price stagnation	Exclude equities with excessive price inactivity
Feature generation feasibility	Verification that engineered features can be reliably constructed	Ensure downstream model compatibility
Predictive feasibility	Logistic Regression under time aware validation	Assess minimum modeling feasibility

Notes: The screening procedure combines data integrity diagnostics, feature generation feasibility checks, and a preliminary predictive feasibility assessment. Numerical thresholds are reported in Appendix A.

4.2 Target Definition and Time Safe Evaluation Protocol

The prediction task is formulated as a binary classification problem representing the direction of the adjusted final price on the following trading day. For each equity and trading day t , the target variable is defined as

$$y_t = \begin{cases} 1, & \text{if } Final_{t+1} > Final_t, \\ 0, & \text{otherwise,} \end{cases}$$

where $Final_t$ denotes the adjusted final price on trading day t . The final price is used as the reference price for defining the prediction target and is distinct from the adjusted closing price associated with the last transaction of the trading session. Accordingly, information available through trading day t is used to predict the direction of the adjusted final price on trading day $t + 1$.

For each equity, observations were ordered chronologically and divided into non overlapping training, validation, and internal test segments. Approximately 63% of the observations were assigned to the training segment, 7% to the validation segment, and the remaining 30% to the internal test segment. Because the available trading histories differ across equities, the exact calendar boundaries of these segments may vary across securities while the same chronological partitioning protocol is maintained.

The training segment was used for model fitting and hyperparameter selection through time aware cross validation. The subsequent validation segment was used to select the classification threshold after model specification. For each fitted model, the decision threshold was selected on the validation observations by maximizing the

Youden criterion. This threshold was then fixed before evaluation on the internal test segment.

The internal test segment was not used for model fitting, hyperparameter tuning, classification threshold selection, or attribution based determination of the dominant feature family. Model performance on this segment therefore provides an out of sample evaluation conditional on the preceding eligibility screening and model development procedure. All feature construction and preprocessing operations preserved chronological ordering, with model dependent preprocessing parameters estimated from the training data rather than from subsequent validation or test observations.

4.3 Machine Learning Models

The empirical framework incorporates four supervised learning algorithms representing complementary learning paradigms: Logistic Regression, Random Forest, XGBoost, and CatBoost. Together, these models provide linear, bagging, and boosting based approaches that are widely used in financial prediction research.

Table 3 summarizes the machine learning models and their respective roles within the empirical framework.

Table 3: Machine Learning Models Used in the Empirical Framework

Model	Learning Paradigm	Primary Role in the Framework
Logistic Regression	Linear	Baseline benchmark and preliminary feasibility assessment
Random Forest	Bagging ensemble	Reference model for permutation importance, feature family aggregation, and ablation analysis
XGBoost	Boosting ensemble	Evaluation of predictive performance across nonlinear boosting models
CatBoost	Boosting ensemble	Evaluation of predictive performance across nonlinear boosting models

Notes: The empirical framework evaluates four supervised learning algorithms representing linear, bagging, and boosting based approaches. Random Forest serves as the reference model for permutation importance, feature family aggregation, and ablation analysis, while all models are evaluated under the same time safe validation protocol.

Logistic Regression was included as a linear benchmark model using balanced class weighting and L2 regularization. Random Forest was selected as the common reference model for permutation based feature attribution and the subsequent feature family ablation analysis. XGBoost and CatBoost were included to extend the analysis to flexible nonlinear boosting models and to examine the consistency of feature family attribution across alternative model specifications.

Model specific preprocessing was implemented within the training pipeline so that data dependent preprocessing parameters were estimated from the corresponding training observations. Hyperparameter optimization was performed exclusively within

the training segment using randomized search with five fold time series cross validation. Following hyperparameter selection, classification thresholds were determined from the validation segment as described in Section 4.2 and were held fixed during internal test evaluation.

4.4 Explainability Analysis

The explainability analysis combines permutation importance and SHAP based attribution analysis to evaluate feature importance using complementary explainability methods. Permutation importance was computed using a fixed reference Random Forest model fitted exclusively on the training observations. Individual features were repeatedly permuted and the resulting change in ROC AUC was used to quantify their importance. Ten permutation repetitions were performed for each feature. When the training segment exceeded 1,200 observations, the most recent 1,200 training observations were used for permutation importance while preserving their temporal order.

Feature level permutation importance values were aggregated according to the predefined feature family taxonomy. For each equity, importance values belonging to the same feature family were summed and subsequently normalized by the total importance across families to obtain family level importance shares. The family with the largest normalized permutation importance share was identified as the dominant feature family for that equity.

SHAP based attribution analysis was conducted using the model specification selected during the preceding validation based model selection procedure. For each equity, the selected model and hyperparameters were reconstructed and the model was fitted exclusively on the training observations. SHAP values were then computed on the transformed training observations, and feature importance was represented by the mean absolute SHAP value of each feature. These feature level values were aggregated and normalized at the family level using the same procedure applied to permutation importance.

To evaluate cross method consistency, the normalized feature family importance structures obtained from permutation importance and SHAP attribution were compared at the aggregate level. Mean family importance shares across equities were normalized within each attribution method and ranked from highest to lowest importance. Rank dispersion across the two methods was then evaluated using the standard deviation of the corresponding family ranks. This procedure provides a direct assessment of whether the relative ordering of feature families is consistent across the two attribution approaches.

Importantly, the internal test segment was not used to estimate feature importance or to identify the dominant feature family. Dominant family identification was based on train only permutation importance and was completed before the subsequent out of sample ablation analysis.

Table 4 summarizes the main components of the explainability analysis and their respective roles within the empirical analysis.

Table 4: Explainability Components Used in the Empirical Framework

Explainability Component	Primary Purpose
Permutation Importance	Quantify predictive degradation after feature permutation and provide the reference importance estimates for feature family aggregation
SHAP Attribution	Estimate additive feature contributions and provide a complementary attribution basis for cross method comparison
Rank Consistency Analysis	Evaluate agreement between permutation based and SHAP based feature family rankings
Family Size Robustness	Evaluate whether family level attribution patterns persist after accounting for unequal numbers of represented features across families

Notes: Explainability is implemented through complementary attribution methods followed by feature family aggregation, cross method consistency analysis, and a family size robustness assessment. All importance estimates are computed after model training using the predefined feature family taxonomy.

4.5 Feature Family Size Robustness

Because the predefined feature families contain different numbers of engineered features, an additional robustness analysis was conducted to examine whether the observed family level attribution structure could be influenced by unequal family sizes. This analysis was applied separately to the permutation importance and SHAP attribution results.

For each equity and attribution method, feature level importance values were first aggregated within each feature family. The resulting raw family importance was then divided by the number of features from that family actually represented in the corresponding attribution output:

$$\bar{I}_{i,f}^{(m)} = \frac{I_{i,f}^{(m)}}{N_{i,f}^{(m)}}, \quad (1)$$

where $I_{i,f}^{(m)}$ denotes the aggregated raw importance of family f for equity i under attribution method m , and $N_{i,f}^{(m)}$ denotes the number of features from that family represented in the corresponding attribution output.

The resulting mean importance per feature was subsequently normalized within each equity and attribution method:

$$S_{i,f}^{(m)} = \frac{\bar{I}_{i,f}^{(m)}}{\sum_g \bar{I}_{i,g}^{(m)}}, \quad (2)$$

where $S_{i,f}^{(m)}$ represents the family size adjusted attribution share. The adjusted dominant family was defined as the family with the largest adjusted share for each equity.

This procedure provides a sensitivity analysis of the primary family level attribution results by examining whether the observed family ranking and dominance patterns persist after accounting for differences in the number of represented features. The family size adjustment is used exclusively as a robustness analysis and does not replace the primary aggregation procedure or alter the dominant family definitions used in the subsequent ablation and trading analyses.

4.6 Dominant versus Non Dominant Ablation

To evaluate whether attributional dominance translates into standalone predictive performance, an out of sample ablation procedure was conducted. The procedure compares predictive performance across alternative feature family configurations for each equity. For each evaluated equity, three feature configurations were constructed: the complete feature set (ALL), the dominant feature family alone (DOM), and all features excluding the dominant family (NONDOM).

Table 5 summarizes the evaluated feature configurations used throughout the ablation analysis.

Table 5: Feature Configurations Used in the Ablation Analysis

Configuration	Description
ALL	Complete engineered feature space used as the full reference configuration.
DOM	Only features belonging to the dominant feature family identified for each equity using train only permutation importance.
NONDOM	All remaining engineered features after excluding the dominant feature family for the corresponding equity.

Notes: The dominant feature family is determined independently for each equity using normalized family level permutation importance computed on the training data only. The family size robustness adjustment does not alter these dominant family assignments. The same ALL, DOM, and NONDOM feature configurations are subsequently used throughout the predictive ablation, statistical inference, and trading based validation analyses.

The dominant feature family for each equity was determined using normalized feature family permutation importance scores obtained from the train only explainability stage. The ablation analysis then compared predictive performance across the ALL, DOM, and NONDOM configurations on the internal test segment. To ensure comparability across feature configurations, a fixed Random Forest model was used throughout the ablation experiments without additional hyperparameter tuning. This fixed model specification was used to isolate differences associated with feature configuration rather than differences arising from repeated model optimization.

The resulting comparisons distinguish the attributional dominance of a feature family from the predictive performance retained when that family is evaluated in isolation or removed from the full feature space. Statistical inference for the predictive evaluation is described in the following subsection.

4.7 Predictive Statistical Inference

To evaluate whether the observed differences in predictive performance are systematic across equities, paired statistical inference was conducted using the equity level evaluation results obtained from the ablation analysis.

Paired Wilcoxon signed rank tests were performed for DOM versus NONDOM, DOM versus ALL, and ALL versus NONDOM. Statistical inference was conducted separately for ROC AUC and PR AUC evaluation metrics to assess predictive performance under complementary discrimination measures.

False discovery rate adjustment using the Benjamini–Hochberg procedure was applied to control for multiple hypothesis testing. In addition to statistical significance, the statistical analysis reports median paired differences, mean paired differences, effect directionality, and sample sizes for each comparison. The equity level comparisons may retain cross sectional dependence arising from common market conditions; accordingly, the resulting significance tests are interpreted as cross sectional paired evidence rather than as evidence based on strictly independent securities.

Table 6 summarizes the statistical inference procedures used throughout the predictive evaluation.

Table 6: Statistical Inference Framework Used in the Empirical Analysis

Statistical Component	Implementation
Paired significance testing	Wilcoxon signed rank test applied to paired equity level performance measures.
Multiple testing correction	Benjamini–Hochberg false discovery rate adjustment.
Evaluation metrics	ROC AUC and PR AUC.
Reported statistics	Mean paired difference, median paired difference, effect directionality, adjusted p value, and sample size.

Notes: Statistical inference is performed using paired equity level comparisons among the DOM, NONDOM, and ALL feature configurations. Multiple testing is controlled using the Benjamini–Hochberg procedure. Statistical inference is conducted separately for ROC AUC and PR AUC.

4.8 Trading Based Validation Procedure

To complement the predictive and explainability analyses, the study also conducts a trading based validation procedure. The purpose of this stage is not to optimize an investment strategy, but to examine whether the predictive differences observed across the ALL, DOM, and NONDOM feature configurations are also reflected in a subsequent out of sample trading evaluation. The out of sample trading evaluation covers the period from November 22, 2025, to June 21, 2026.

The validation uses the previously fitted models and their associated classification thresholds for the three feature configurations introduced earlier: ALL, DOM, and NONDOM. For each equity, the ALL configuration uses the full feature space, the DOM configuration uses only the features belonging to the dominant feature

family of that equity, and the NONDOM configuration uses all remaining features after excluding the dominant feature family. The model specifications and classification thresholds are held fixed throughout the trading period, with no additional model fitting, hyperparameter tuning, or threshold optimization. Each configuration is evaluated separately at the individual equity level, and no cross equity portfolio is constructed. This design preserves the central objective of the study by evaluating the three feature configurations under a common subsequent trading procedure rather than optimizing a trading strategy.

Trading signals are generated using only information available through each signal date, consistent with the time safe evaluation protocol introduced earlier. Specifically, feature values observed at day t are passed to the corresponding model to estimate the probability that the adjusted final price at day $t + 1$ exceeds that at day t . A positive signal is generated when the predicted probability is greater than or equal to the previously selected classification threshold; otherwise, the signal is zero. Because the prediction signal is formed using information from the completed trading session at day t , the adjusted opening price of day $t + 1$ is used as the subsequent execution price. Returns are therefore computed using an open to open convention.

The trading rule is intentionally simple and deterministic. If the model generates a positive signal while the equity is not currently held, a long position is opened at the next available opening price. If the equity is already held and the signal remains positive, the position is maintained. If the signal switches to zero, the position is closed at the next opening price. Because short selling is not permitted in the examined market environment, the strategy only allows long or flat positions. Transaction costs are incorporated at rates of 0.3712% for purchases and 0.8800% for sales. The reported performance measures therefore reflect trading outcomes after these transaction costs, but before market impact, order book constraints, and live execution uncertainty.

The trading evaluation was restricted to equities with adjusted OHLCV data extending through the complete required validation period. Of the 250 equities in the modeling universe, 231 satisfied this requirement and were successfully evaluated. The remaining 19 equities were excluded because their available price histories ended before the required end date of the trading validation period.

For each equity and feature configuration, the validation records total return, annualized return, annualized volatility, Sharpe ratio, exposure, number of trades, win rate, profit factor, and holding period statistics. The empirical discussion focuses on total return and Sharpe ratio because these measures capture raw trading performance and risk adjusted performance, respectively. The resulting outputs are used as an additional validation layer for the feature family framework rather than as evidence of an optimized investment strategy.

4.9 Clustering and Industry Overlap Analysis

To explore whether similarities in feature family importance profiles correspond to industry classifications, a clustering based analysis was conducted using feature family importance profiles obtained from the train only permutation importance analysis.

For each equity, a normalized feature family importance vector was constructed from the aggregated permutation importance scores. Hierarchical clustering was then

applied to these vectors as an exploratory procedure to identify groups of equities exhibiting similar feature family importance profiles. Alternative clustering solutions were evaluated using the silhouette coefficient and Davies–Bouldin index, and the selected clustering solution was used for the subsequent industry overlap analysis.

The clustering analysis was conducted independently of the predictive modeling and ablation stages and did not affect model selection, dominant family identification, or out of sample evaluation. Its purpose was limited to examining whether similarities in the estimated feature family importance profiles exhibit observable grouping patterns.

To assess the correspondence between the resulting data driven clusters and conventional market classifications, cluster assignments were compared with industry labels associated with TSE sector categories. Agreement between the two classifications was evaluated using Normalized Mutual Information (NMI) and the Adjusted Rand Index (ARI). This analysis was used to examine whether equities with similar feature family importance profiles also exhibit correspondence with industry membership, without treating the absence of industry alignment as evidence of a market wide structure.

4.10 Implementation and Reproducibility Details

All analyses were implemented in Python using a consistent computational workflow. Model estimation, feature engineering, explainability analysis, ablation analysis, and statistical inference were conducted using chronologically ordered data and the predefined empirical procedures described in the preceding sections. Randomized procedures were controlled using fixed random seeds where applicable.

Temporal ordering was preserved throughout model development and evaluation. Preprocessing parameters were estimated within the corresponding training data, and the predefined training, validation, and test roles were maintained throughout the predictive analysis. The same feature construction rules and feature family definitions were applied consistently across the empirical analyses.

To preserve comparability, the methodological settings defined for each stage of the analysis were applied consistently across equities and the corresponding feature configurations. Model fitting, explainability estimation, dominant family identification, ablation analysis, and statistical evaluation followed the time safe procedures described in the preceding subsections.

To facilitate transparency and reproducibility, the analysis notebook and supporting empirical outputs used in the reported analyses are provided as Online Resource 1. The supplementary materials document the computational workflow and provide the saved outputs underlying the principal empirical results reported in the manuscript.

5 Empirical Results

5.1 Predictive Performance Across Model Families

All four evaluated models achieve median ROC AUC and PR AUC values above 0.5 on the internal test segment under the time ordered evaluation protocol. Differences

across model families are relatively modest, with Random Forest exhibiting the highest median performance on both evaluation metrics.

Random Forest achieves a median ROC AUC of 0.7136 and a median PR AUC of 0.6909 across the 250 evaluated equities. Logistic Regression follows with median values of 0.7041 and 0.6797, respectively. XGBoost achieves a median ROC AUC of 0.6926 and a median PR AUC of 0.6737, while the corresponding values for CatBoost are 0.6860 and 0.6639. These results indicate that greater model complexity does not systematically correspond to stronger predictive performance within the examined setting.

Random Forest serves as the common reference model for the subsequent permutation based feature attribution and feature family ablation analyses, as specified in Section 4.3.

Table 7 summarizes predictive performance across the evaluated model families using ROC AUC and PR AUC measured on the internal test segment.

Table 7: Cross Sectional Predictive Performance Across Machine Learning Models

Model	Equities	Median ROC AUC	IQR ROC AUC	Median PR AUC	IQR PR AUC
Logistic Regression	250	0.7041	0.0497	0.6797	0.0761
Random Forest	250	0.7136	0.0527	0.6909	0.0693
XGBoost	250	0.6926	0.0615	0.6737	0.0825
CatBoost	250	0.6860	0.0540	0.6639	0.0882

Notes: Predictive performance is evaluated using out of sample test set observations. Reported values summarize the cross sectional distribution of model performance across the 250 evaluated equities. IQR denotes the interquartile range.

5.2 Feature Family Dominance

Feature level permutation importance values obtained from the train only reference Random Forest model were aggregated into the feature family framework introduced in Section 3. The resulting feature family importance distributions were then evaluated across the full modeling universe.

Table 8 summarizes the cross sectional feature family importance structure using mean family share, median family share, and dominant family frequency.

The results reveal a concentrated family level importance structure. The Daily Price Structure family exhibits the highest mean importance share (38.56%) and median importance share (38.88%) across equities. Momentum represents the second highest family, with a mean share of 25.37% and a median share of 24.77%. Volume flow and liquidity features account for a mean importance share of 20.40%, followed by volatility related features (9.62%) and trend related features (6.05%).

The dominance frequency results show a similarly concentrated pattern. Daily Price Structure emerges as the dominant feature family for 85.60% of the evaluated

equities (214 of 250). In comparison, momentum is dominant for 13.20% of equities (33 of 250), while volume flow and liquidity is dominant for 1.20% (3 of 250). Neither the volatility nor the trend feature family emerges as dominant for any evaluated equity.

Aggregate importance shares and dominance frequencies provide complementary descriptions of the family level attribution structure. Momentum and volume flow and liquidity retain meaningful aggregate importance shares, while Daily Price Structure most frequently exhibits the largest family level permutation importance share at the individual equity level.

Trend exhibits the lowest aggregate importance share among the five feature families and does not emerge as dominant for any evaluated equity. Volatility accounts for a larger but still comparatively modest share of aggregate importance and likewise does not emerge as the dominant family for any evaluated equity.

Overall, the permutation importance results indicate that family level attribution within the examined modeling universe is concentrated primarily in Daily Price Structure. This pattern is observed in both aggregate importance shares and equity level dominance frequencies.

Figure 2 illustrates the distribution of dominant feature families across equities.

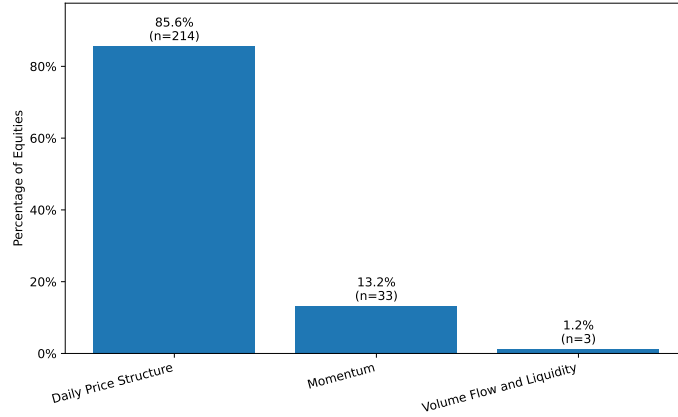


Fig. 2: Distribution of Dominant Feature Families Across Equities

5.3 Cross Method Explainability Agreement

Feature family attribution patterns obtained from train only permutation importance and SHAP based attribution were compared to evaluate cross method consistency. Feature level importance values from both attribution methods were aggregated into the predefined feature families introduced in Section 3 and normalized within each method.

Table 9 summarizes the feature family importance distributions derived from permutation importance and SHAP attribution.

Table 8: Feature Family Importance Structure Across Equities

Feature Family	Mean Share (%)	Median Share (%)	Dominant Frequency (%)
Daily Price Structure	38.56	38.88	85.60
Momentum	25.37	24.77	13.20
Volume Flow and Liquidity	20.40	20.35	1.20
Volatility	9.62	9.36	0.00
Trend	6.05	5.95	0.00

Notes: Mean share and median share summarize the cross sectional distribution of normalized family level permutation importance across equities. Dominant frequency reports the percentage of equities for which a given family has the largest normalized family level permutation importance share.

The two attribution methods produce identical feature family rankings. Under both methods, Daily Price Structure ranks first, followed by momentum, volume flow and liquidity, volatility, and trend. Daily Price Structure therefore receives the largest aggregate attribution share under both methods, while momentum and volume flow and liquidity occupy the second and third positions, respectively.

The absolute family importance shares differ between permutation importance and SHAP attribution, but these differences do not alter the relative ordering of the five feature families. Thus, the observed cross method consistency is most clearly reflected in the stability of the family rankings rather than in equality of attribution magnitudes.

To quantify ranking consistency, rank dispersion was evaluated across feature families using permutation importance and SHAP based rankings. The resulting rank standard deviation is zero for all five feature families, indicating complete rank agreement between the two attribution methods.

This complete rank agreement indicates that the relative ordering of feature families is not specific to either of the two attribution methods examined. In particular, the position of Daily Price Structure as the highest ranked family is reproduced by both permutation importance and SHAP attribution.

Table 9: Cross Method Family Importance Comparison

Feature Family	Permutation Share (%)	SHAP Share (%)	Permutation Rank	SHAP Rank
Daily Price Structure	38.56	29.66	1	1
Momentum	25.37	29.38	2	2
Volume Flow and Liquidity	20.40	20.66	3	3
Volatility	9.62	11.98	4	4
Trend	6.05	8.33	5	5

Notes: Family level importance shares were obtained from train only permutation importance and SHAP attribution. Both attribution methods produce identical family rankings, indicating complete agreement in the relative family ordering across the two attribution methods.

Figure 3 compares permutation based and SHAP based feature family importance estimates and illustrates the consistency of the resulting family rankings.

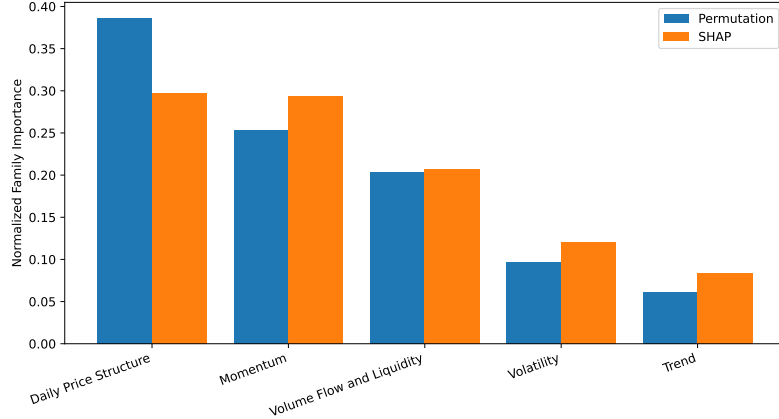


Fig. 3: Cross Method Comparison of Feature Family Importance Rankings

5.4 Feature Family Size Robustness

Because the feature families contain different numbers of engineered features, an additional robustness analysis was conducted to examine whether the primary family level attribution patterns persist after accounting for unequal family sizes. As described in Section 4, the adjustment is based on the mean importance per represented feature within each family and is performed separately for permutation importance and SHAP attribution.

Table 10 reports the resulting family size adjusted attribution shares and adjusted dominant family frequencies.

The family size adjustment does not weaken the prominence of Daily Price Structure. Under permutation importance, Daily Price Structure has a mean adjusted attribution share of 47.57% and a median adjusted share of 47.87%. It is identified as the adjusted dominant family for 244 of the 250 evaluated equities (97.6%). The corresponding mean adjusted shares are 15.08% for Momentum, 14.48% for Volume Flow and Liquidity, 12.41% for Volatility, and 10.99% for Trend.

The SHAP based adjustment produces the same central pattern. Daily Price Structure retains the largest mean adjusted attribution share at 38.10% and the largest median adjusted share at 36.98%, and it is identified as the adjusted dominant family for 222 of the 250 equities (88.8%). The corresponding mean adjusted shares are 17.54% for Momentum, 15.70% for Volume Flow and Liquidity, 15.63% for Volatility, and 13.19% for Trend.

These results indicate that the prominence of Daily Price Structure observed in the primary attribution analysis is not explained by its number of constituent features. After accounting for unequal family sizes, Daily Price Structure remains the family

Table 10: Feature Family Size Robustness of Attribution Results

Feature Family	Permutation Importance			SHAP		
	Mean Adjusted Share (%)	Median Adjusted Share (%)	Adjusted Domi- nant (%)	Mean Adjusted Share (%)	Median Adjusted Share (%)	Adjusted Domi- nant (%)
Daily Price Structure	47.57	47.87	97.6	38.10	36.98	88.8
Momentum	15.08	14.05	2.0	17.54	16.74	4.4
Volume Flow and Liquidity	14.48	13.97	0.4	15.70	15.82	1.2
Volatility	12.41	12.05	0.0	15.63	15.03	4.8
Trend	10.99	10.37	0.0	13.19	12.91	0.8

Notes: Adjusted shares are obtained by dividing raw family importance by the number of features from that family represented in the corresponding attribution output and then renormalizing within each equity. Mean and median shares are based on 250 equities for all families except Trend, for which 238 permutation importance observations and 247 SHAP observations are available. Adjusted dominant percentages are calculated across the full set of 250 equities.

with the largest average adjusted attribution share under both permutation importance and SHAP and remains the adjusted dominant family for the large majority of evaluated equities.

The family size analysis is interpreted as a sensitivity assessment of the primary attribution results. It does not redefine the dominant families used in the subsequent ALL, DOM, and NONDOM analyses, which remain based on the original train only permutation importance aggregation.

5.5 Dominant versus Non Dominant Ablation Analysis

To evaluate whether attributional dominance is accompanied by standalone predictive performance, an out of sample ablation analysis was conducted using the three feature configurations defined earlier: ALL, DOM, and NONDOM.

Table 11 summarizes the cross sectional predictive performance of the three configurations.

Table 11: Overall Ablation Performance Across Feature Configurations

Configuration	Mean ROC AUC	Mean PR AUC
ALL	0.6992	0.6754
DOM	0.7009	0.6736
NONDOM	0.6089	0.5933

Notes: Reported values summarize mean out of sample predictive performance across the evaluated equities. DOM exhibits the highest mean ROC AUC, whereas ALL exhibits the highest mean PR AUC. NONDOM shows substantially lower mean performance under both evaluation metrics.

Across the full modeling universe, the DOM and ALL configurations exhibit similar average predictive performance, whereas the NONDOM configuration performs substantially lower on both evaluation metrics. Mean ROC AUC is 0.6992 for ALL, 0.7009 for DOM, and 0.6089 for NONDOM. Mean PR AUC is 0.6754 for ALL, 0.6736 for DOM, and 0.5933 for NONDOM. Thus, evaluating the dominant family in isolation results in performance close to that of the complete feature set at the aggregate level, while excluding the dominant family is associated with a larger reduction in predictive performance.

To examine whether this aggregate pattern varies according to the identity of the dominant family, the ablation results were further stratified by the dominant feature family assigned to each equity. Table 12 reports the corresponding results.

Table 12: Ablation Performance Stratified by Dominant Family

Dominant Family	Equities	ALL ROC AUC	DOM ROC AUC	NONDOM ROC AUC
Daily Price Structure	214	0.6999	0.7113	0.5951
Momentum	33	0.6947	0.6475	0.6886
Volume Flow and Liquidity	3	0.6992	0.5450	0.7172

Notes: Reported values summarize mean ROC AUC within equities grouped according to their dominant feature family. Daily Price Structure dominated equities exhibit higher mean ROC AUC under DOM than under ALL and NONDOM, whereas the momentum dominated subgroup exhibits lower mean performance under DOM than under ALL and NONDOM. Results for Volume Flow and Liquidity should be interpreted cautiously because this subgroup contains only three equities.

Among the 214 equities dominated by Daily Price Structure, the DOM configuration achieves a mean ROC AUC of 0.7113, compared with 0.6999 for ALL and 0.5951 for NONDOM. In contrast, among the 33 momentum dominated equities, DOM achieves a mean ROC AUC of 0.6475, compared with 0.6947 for ALL and 0.6886 for NONDOM. For the three equities dominated by Volume Flow and Liquidity, the corresponding values are 0.5450 for DOM, 0.6992 for ALL, and 0.7172 for NONDOM. Given the very small size of the Volume Flow and Liquidity subgroup, these values are reported descriptively and are not used to support broader conclusions about that family.

These subgroup results demonstrate an important distinction between attributional dominance and standalone predictive performance. A family identified as the largest attribution contributor does not necessarily reproduce the predictive performance of the complete feature set when evaluated in isolation. Within the examined universe, standalone performance of the DOM configuration is concentrated primarily among equities dominated by Daily Price Structure, whereas the momentum dominated subgroup exhibits a different pattern.

Table 13 reports the paired statistical comparisons among the three ablation configurations.

Paired Wilcoxon signed rank tests indicate statistically significant differences between DOM and NONDOM and between ALL and NONDOM for both ROC AUC

Table 13: Paired Statistical Inference for Ablation Performance

Comparison	Median Paired Difference	FDR Adjusted p Value
DOM vs NONDOM (ROC AUC)	+0.1036	< 0.001
DOM vs ALL (ROC AUC)	+0.0066	0.0333
ALL vs NONDOM (ROC AUC)	+0.0922	< 0.001
DOM vs NONDOM (PR AUC)	+0.0956	< 0.001
DOM vs ALL (PR AUC)	+0.0016	0.8370
ALL vs NONDOM (PR AUC)	+0.0853	< 0.001

Notes: Paired Wilcoxon signed rank tests compare the DOM, NONDOM, and ALL feature configurations across the 250 evaluated equities. Differences are calculated as the first listed configuration minus the second listed configuration. Reported p values are adjusted using the Benjamini–Hochberg false discovery rate procedure.

and PR AUC after Benjamini–Hochberg false discovery rate adjustment. For DOM versus ALL, the ROC AUC comparison is statistically significant, with a median paired difference of +0.0066 and an adjusted p value of 0.0333, whereas the PR AUC comparison is not statistically significant, with a median paired difference of +0.0016 and an adjusted p value of 0.8370. The small magnitude of the DOM versus ALL differences indicates that statistical significance in the ROC AUC comparison should be distinguished from the practical magnitude of the observed performance difference.

Figure 4 summarizes the overall out of sample performance of the three ablation configurations.

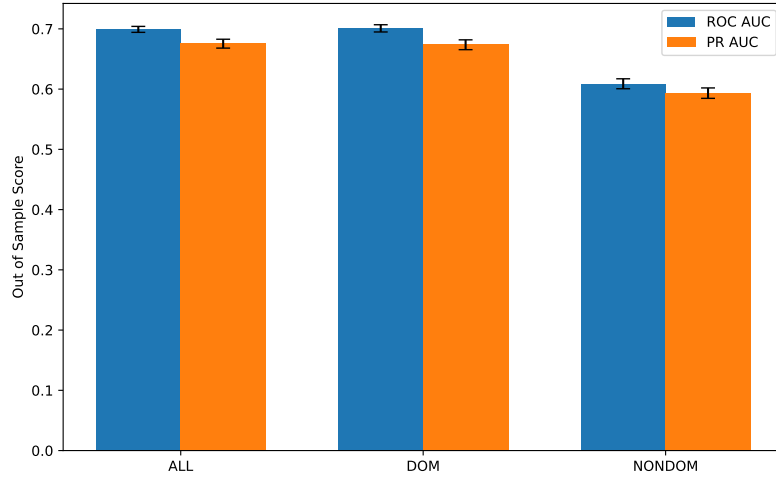


Fig. 4: Out of Sample Performance Under Alternative Ablation Configurations. Error bars denote ± 1.96 standard errors around the cross sectional mean.

Taken together, the ablation results show that attributional dominance and standalone predictive performance are related but distinct properties. At the aggregate level, DOM and ALL exhibit similar predictive performance while NONDOM performs substantially lower. The stratified results further show that this aggregate pattern is driven predominantly by the large Daily Price Structure dominant subgroup and should not be generalized uniformly across all dominant feature families.

5.6 Clustering Structure and Industry Overlap

To examine whether feature family importance profiles exhibit systematic alignment with industry classifications, a clustering analysis was conducted using feature family importance profiles obtained from the train only permutation importance method. Each equity was represented by its normalized feature family importance vector, and hierarchical clustering was applied to identify groupings within the resulting feature family importance space.

Table 14 summarizes the clustering diagnostics for the selected clustering solution. The results provide limited evidence of distinct segmentation within the feature family importance space. The clustering solution is highly concentrated, with 238 of the 250 evaluated equities (95.2%) assigned to one cluster and the remaining 12 equities (4.8%) assigned to the second cluster.

Industry overlap diagnostics show weak correspondence between the clustering assignments and conventional industry classifications. Both the normalized mutual information (NMI) and adjusted Rand index (ARI) are close to zero. Consistent with these results, the industry enrichment analysis does not identify statistically significant industry concentration after multiple testing adjustment.

Taken together, these results provide little evidence that the feature family importance profiles in the examined modeling universe are systematically partitioned according to conventional industry classifications. Given the highly imbalanced clustering solution and the weak correspondence with industry labels, the clustering analysis does not support stronger conclusions regarding the existence of distinct industry based predictive structures.

Table 14: Clustering Diagnostics and Industry Overlap

K	Silhouette	Davies–Bouldin	NMI vs Industry	ARI vs Industry
2	0.3632	0.9349	0.0476	0.0034

Notes: Hierarchical clustering was performed using normalized feature family importance profiles. NMI and ARI quantify the correspondence between cluster assignments and industry classifications. NMI and ARI values close to zero indicate weak correspondence between the two structures. The selected clustering solution partitions the evaluated equities into two clusters containing 238 and 12 equities, respectively.

5.7 Integrated Robustness Assessment

The empirical analyses provide complementary evidence regarding different aspects of the identified feature family importance structure. Rather than treating each analysis as an independent confirmation of a single attribution structure, the results are considered jointly to assess the consistency of family level attribution patterns and their relationship with subsequent predictive performance.

The first source of evidence is the cross method explainability analysis. Permutation importance and SHAP attribution produce identical rankings across all five feature families. Although the absolute attribution shares differ between methods, the complete rank agreement indicates that the relative ordering of the feature families is consistent across the two attribution approaches examined.

The family size robustness analysis addresses a separate potential influence on the attribution results arising from unequal numbers of features across the predefined families. After adjusting family attribution for the number of represented features, Daily Price Structure retains the largest average adjusted attribution share under both permutation importance and SHAP and remains the adjusted dominant family for the large majority of evaluated equities. The prominence of Daily Price Structure in the primary attribution analysis is therefore not explained by unequal family sizes.

The ablation analysis provides a complementary predictive evaluation of attributional dominance. The DOM and ALL configurations exhibit similar aggregate predictive performance, whereas the NONDOM configuration performs substantially lower on both ROC AUC and PR AUC. However, the stratified analysis shows that attributional dominance does not translate uniformly into standalone predictive performance across dominant feature families. The aggregate pattern is driven predominantly by the large Daily Price Structure dominant subgroup, while the momentum dominated subgroup exhibits weaker standalone performance under the DOM configuration.

Formal statistical inference further characterizes these performance differences. Paired Wilcoxon signed rank tests indicate statistically significant differences between DOM and NONDOM and between ALL and NONDOM for both ROC AUC and PR AUC after false discovery rate adjustment. For DOM versus ALL, the ROC AUC difference is statistically significant, whereas the PR AUC difference is not. Importantly, the observed DOM versus ALL differences are small in magnitude, indicating that statistical significance in the ROC AUC comparison should not be interpreted as a large practical performance advantage.

The clustering analysis addresses a different question. The resulting clusters exhibit weak correspondence with conventional industry classifications, and the industry enrichment analysis does not identify statistically significant industry concentration after multiple testing adjustment. These results provide little evidence that the observed feature family importance profiles are systematically partitioned according to conventional industry categories. They do not, however, establish that the identified structure constitutes a market wide phenomenon.

Taken together, the results show that the relative family ranking is consistent across the two attribution methods and that the prominence of Daily Price Structure persists after accounting for unequal family sizes. The ablation results further show

that the DOM and ALL configurations exhibit similar aggregate predictive performance, whereas excluding the dominant family under the NONDOM configuration is associated with substantially lower performance. At the same time, standalone predictive performance varies across dominant families. The clustering analysis places a separate boundary on structural interpretation: the absence of industry alignment does not by itself imply a common market wide structure. The trading based validation presented in the following section provides an additional evaluation of the ALL, DOM, and NONDOM configurations in a subsequent out of sample period.

5.8 Trading Based Validation Results

Although the primary objective of the present study is to investigate the organization of predictive information rather than to develop a trading strategy, an additional trading based validation was conducted to compare the ALL, DOM, and NONDOM configurations during a subsequent out of sample period. This analysis provides a descriptive assessment of whether differences among the three feature configurations are also observable in trading based outcomes.

Table 15 reports the trading based validation results in two panels. Panel A summarizes performance across all 231 successfully evaluated equities, whereas Panel B restricts the analysis to the 203 equities whose dominant feature family is Daily Price Structure within the trading validation sample.

In Panel A, the DOM configuration exhibits the highest median total return and median Sharpe ratio. Median total return is 8.01% under ALL, 13.38% under DOM, and 9.54% under NONDOM. The corresponding median Sharpe ratios are 0.91, 1.24, and 1.14, respectively. Thus, DOM exhibits the strongest median trading outcomes across the full evaluated trading sample, while NONDOM lies between DOM and ALL on both reported measures.

Panel B shows a similar descriptive ordering among the 203 Daily Price Structure dominated equities. Median total return is 6.35% for ALL, 13.02% for DOM, and 7.95% for NONDOM, while the corresponding median Sharpe ratios are 0.79, 1.13, and 0.92. Within this subgroup, DOM therefore again exhibits the highest median total return and median Sharpe ratio.

These descriptive patterns are consistent with the standalone performance of the DOM configuration previously observed for Daily Price Structure dominated equities in the predictive ablation analysis. However, the trading results represent a separate evaluation based on trading outcomes and should not be interpreted as a direct measure of predictive information or as statistical confirmation of differences among the three configurations.

Overall, the trading based validation provides complementary descriptive evidence regarding the behavior of the ALL, DOM, and NONDOM configurations in a subsequent out of sample period. The results are not intended to establish an optimized investment strategy or statistically significant trading superiority and should not be interpreted as evidence of deployable trading profitability.

Table 15: Trading Based Performance Across Feature Configurations

Sample	Config.	Equities	Median Total Return (%)	IQR Total Return (%)	Median Sharpe
<i>Panel A: All evaluated equities</i>					
All evaluated equities	ALL	231	8.01	37.20	0.91
All evaluated equities	DOM	231	13.38	36.24	1.24
All evaluated equities	NONDOM	231	9.54	32.33	1.14
<i>Panel B: Daily Price Structure dominant equities</i>					
Daily Price Structure dominant equities	ALL	203	6.35	32.16	0.79
Daily Price Structure dominant equities	DOM	203	13.02	37.79	1.13
Daily Price Structure dominant equities	NONDOM	203	7.95	32.62	0.92

Notes: Reported values summarize cross sectional trading performance across successfully evaluated equities. Panel A reports results for the full evaluated backtest sample, while Panel B restricts the sample to equities whose dominant feature family is Daily Price Structure. Trading signals are generated using the saved out of sample models under identical trading rules and transaction cost assumptions across feature configurations. Total Return denotes cumulative net trading return after transaction costs, while the Sharpe Ratio summarizes risk adjusted trading performance.

6 Discussion

6.1 Interpreting Daily Price Structure Dominance

One of the central empirical findings of the present study is the prominence of the Daily Price Structure family within the examined modeling universe. This family receives the highest aggregate attribution share under both permutation importance and SHAP and is identified as the dominant family for the large majority of evaluated equities. The ablation results provide a complementary finding: among equities dominated by Daily Price Structure, the DOM configuration retains standalone predictive performance and remains comparable to the full feature configuration. These results distinguish attributional dominance from standalone predictive performance while showing that, for Daily Price Structure dominated equities, the two properties are closely aligned.

The family size robustness analysis provides additional support for the attributional prominence of Daily Price Structure. After adjusting family attribution for the number of represented features within each family, Daily Price Structure retains the largest average adjusted attribution share under both permutation importance and SHAP and remains the adjusted dominant family for the large majority of evaluated equities. This result indicates that its prominent position in the primary attribution analysis is not explained by unequal family sizes.

The prominence of Daily Price Structure suggests that variables describing the configuration of prices within the daily trading period contain substantial predictive relevance for the next period directional target examined in this study. These variables characterize aspects of daily price behavior such as candlestick body magnitude, upper and lower shadows, closing location within the daily range, and the relation

between closing and final prices. Their importance therefore indicates that the internal configuration of daily prices contributes information that is not fully represented by the momentum, volatility, volume flow and liquidity, or trend families included in the feature space.

Several interpretations of this pattern are possible, but they cannot be distinguished directly by the present empirical design. Daily price configurations may reflect short horizon adjustments in trading activity, liquidity conditions, or the incorporation of information into prices. However, the daily OHLCV data used in this study do not directly observe order flow, order book dynamics, investor composition, or transaction level behavior. The observed importance of Daily Price Structure should therefore be interpreted as evidence of predictive relevance in daily price configurations rather than as direct evidence regarding the underlying market mechanisms that generate those configurations.

The ablation analysis further demonstrates that attributional dominance does not translate uniformly into standalone predictive performance across dominant feature families. Daily Price Structure dominated equities exhibit standalone performance under the DOM configuration, whereas the momentum dominated subgroup shows substantially weaker performance when its dominant family is evaluated in isolation. The trading based validation provides a related descriptive result: within the Daily Price Structure dominated trading sample, the DOM configuration exhibits the highest median total return and median Sharpe ratio among the three evaluated configurations. Because the trading comparisons are descriptive, this finding should be viewed as complementary evidence rather than statistical confirmation of trading superiority.

The comparatively low attribution assigned to trend related indicators provides an additional contrast. Within the feature space and prediction horizon examined here, trend features receive the smallest aggregate attribution share and are not identified as the dominant family for any evaluated equity. One possible interpretation is that the slower moving information represented by these indicators is less closely aligned with the short horizon directional target considered in the study. This interpretation remains tentative, however, because the empirical design evaluates predictive relevance rather than the economic mechanisms underlying differences among feature families.

Importantly, these findings should not be interpreted as evidence that Daily Price Structure variables universally dominate financial prediction tasks or market environments. The observed pattern is conditional on the screened modeling universe, daily observation frequency, selected prediction horizon, engineered feature set, model specifications, and evaluation procedures employed in the study.

The identified relationships should likewise not be interpreted causally. The explainability and ablation analyses characterize predictive associations and model dependence on the engineered feature families; they do not establish that changes in Daily Price Structure variables cause subsequent price movements.

Overall, the evidence identifies Daily Price Structure as the most prominent family within the examined feature space while also clarifying the meaning of that prominence. Its position is supported by consistent attribution rankings and standalone predictive performance within the corresponding dominant subgroup, but these findings concern predictive organization rather than causal market mechanisms.

More broadly, the results demonstrate the importance of distinguishing which feature families receive the greatest model attribution from whether those families can independently reproduce the predictive performance of the broader feature space.

6.2 Industry Alignment of Feature Family Importance Profiles

The clustering analysis was motivated by the possibility that equities from different industries might exhibit systematically different feature family importance profiles. The results provide little evidence for such an alignment within the examined modeling universe. The selected clustering solution is highly imbalanced, with 238 of the 250 equities assigned to one cluster and only 12 assigned to the second, while the correspondence between cluster assignments and conventional industry classifications is weak.

The low normalized mutual information and adjusted Rand index values indicate limited agreement between the data driven clustering assignments and industry labels. Consistent with these measures, the industry enrichment analysis does not identify statistically significant industry concentration after multiple testing adjustment. Taken together, these results do not support the hypothesis that the observed feature family importance profiles are strongly organized according to conventional industry classifications.

This finding should be interpreted narrowly. The absence of strong correspondence with industry labels does not imply that industry effects are irrelevant to financial prediction, nor does it establish that the identified feature family importance structure constitutes a market wide phenomenon. Industry membership may remain relevant to return dynamics, firm fundamentals, liquidity conditions, or other dimensions that are not represented by the particular feature family importance profiles examined here.

The results instead indicate that conventional industry classification does not provide a strong explanation for the grouping observed in the feature family importance space. This distinction is important because industry classifications organize firms according to their economic activities, whereas the clustering analysis operates on model derived feature family importance profiles constructed from the engineered daily OHLCV feature space. Strong correspondence between these two forms of classification therefore need not be expected.

The highly imbalanced clustering solution also limits the strength of structural interpretation. With most equities assigned to a single cluster, the analysis provides little evidence of clearly separated subgroups within the feature family importance space. Accordingly, the clustering results are more informative as evidence against pronounced industry based segmentation than as evidence for an alternative form of market organization.

Overall, the clustering analysis does not support the initial possibility that distinct industry groups are associated with clearly differentiated feature family importance profiles within the examined modeling universe. Beyond this conclusion, stronger claims regarding market wide predictive structures or common underlying market mechanisms would require additional evidence and are not inferred from the present clustering results.

6.3 Explainability and Financial Machine Learning

The present study extends the use of explainability in financial machine learning from individual feature attribution toward the empirical investigation of how predictive relevance is organized across interpretable feature families. As machine learning models become increasingly flexible and capable of capturing complex nonlinear relationships, interpretation based solely on individual feature rankings may provide a fragmented representation of model behavior, particularly when related financial variables convey overlapping information. Within this context, feature family aggregation provides a higher level representation through which attribution patterns can be examined across related groups of predictors.

The proposed framework treats explainability as a central analytical component rather than solely as a reporting mechanism. Feature level attribution is first aggregated into predefined families, after which the resulting family level structure is examined across complementary attribution methods. The framework then distinguishes attributional dominance from subsequent predictive performance through the ALL, DOM, and NONDOM ablation configurations. In this way, explainability is used not only to identify which feature families receive greater model attribution, but also to motivate an empirical evaluation of what that dominance implies when the corresponding feature subsets are evaluated separately.

The cross method results provide particularly clear evidence regarding attributional consistency. Train only permutation importance and SHAP produce identical rankings across all five feature families, despite differences in their absolute attribution shares. This agreement indicates that the relative family ordering is consistent across the two attribution approaches examined. Importantly, however, the ablation results show that attributional dominance does not translate uniformly into standalone predictive performance across dominant feature families. Among Daily Price Structure dominated equities, the DOM configuration performs comparably to, and on average above, ALL, whereas the momentum dominated subgroup exhibits weaker standalone performance.

This distinction has broader implications for the interpretation of machine learning models in finance. A feature family may receive the largest share of model attribution without reproducing, in isolation, the predictive performance obtained from the broader feature space. Attributional dominance and standalone predictive performance therefore represent related but conceptually distinct properties. Evaluating them separately can provide a more informative interpretation of model behavior than relying exclusively on attribution rankings.

Feature family aggregation also provides a more interpretable level of analysis than isolated feature rankings within the present setting. Organizing the engineered variables into Daily Price Structure, momentum, volume flow and liquidity, volatility, and trend enables the attribution results to be interpreted in terms of predefined dimensions of the feature space while retaining the underlying feature level information. The purpose of this aggregation is not to eliminate feature level heterogeneity, but to provide a complementary representation of how model attribution is distributed across related groups of variables.

At the same time, feature family aggregation does not resolve the broader interpretability challenges associated with financial machine learning. Attribution methods remain descriptive of model behavior rather than causal mechanisms, and aggregation necessarily depends on the feature taxonomy adopted in the study. The resulting family level structure should therefore be interpreted as an empirical representation conditional on the engineered feature space, family definitions, models, and evaluation procedures used in the analysis.

Overall, the findings illustrate how explainability can be used for more than individual feature ranking. By combining feature family aggregation, cross method attribution comparison, and subsequent ablation analysis, the proposed framework provides a structured approach for examining both the distribution of model attribution and the predictive implications of attributional dominance. This distinction between attributional dominance and standalone predictive performance represents an important component of the study’s interpretation of predictive information organization.

6.4 Practical and Regulatory Implications

The findings of the present study have several implications for financial modeling, model interpretation, and the governance of machine learning systems. These implications should be considered within the scope of the empirical framework. The objective of the study is not to develop a deployable trading strategy or optimize portfolio performance, but to investigate how model attribution and predictive performance are organized across interpretable feature families within the examined setting.

One practical implication concerns the organization and interpretation of financial features. The results show that family level attribution is not distributed uniformly across the engineered feature space and that attributional dominance does not necessarily imply equivalent standalone predictive performance. This distinction suggests that feature evaluation may benefit from considering both the relative attribution assigned to groups of related variables and the predictive behavior of those groups when evaluated separately. Feature family aggregation provides one structured way of conducting such an analysis while retaining the underlying feature level information.

The feature family perspective may also be relevant to model governance and communication. Individual feature explanations can become difficult to summarize when models incorporate numerous related predictors, whereas aggregation into pre-defined and interpretable families can provide a higher level representation of model attribution. Such representations may facilitate communication of model behavior to researchers, model developers, and other stakeholders by showing how attribution is distributed across broader components of the feature space. This potential benefit should not be interpreted as evidence that feature family aggregation by itself satisfies regulatory or governance requirements, which depend on the specific application and institutional context.

A further practical implication concerns the interpretation of feature importance. The ablation results demonstrate that the family receiving the largest attribution share is not necessarily sufficient in isolation to reproduce the performance of the broader feature set. Consequently, decisions regarding feature selection or model simplification should not be based solely on attribution rankings. Evaluating the predictive

consequences of retaining or removing attributed feature groups can provide additional information about the relationship between model attribution and predictive dependence.

These considerations may be particularly relevant when interpretable machine learning is applied to financial environments in which the predictive role of different feature groups is uncertain or heterogeneous. At the same time, the empirical findings of the present study are conditional on the examined modeling universe and should not be generalized to other markets, frequencies, prediction horizons, or feature spaces without further validation.

The trading based validation provides a complementary illustration of the behavior of the ALL, DOM, and NONDOM configurations during a subsequent out of sample period. The higher median total return and Sharpe ratio observed for the DOM configuration, particularly among Daily Price Structure dominated equities, are descriptively consistent with the corresponding ablation results. However, these comparisons were not designed to establish statistically significant trading superiority, and the study does not account for all constraints associated with live implementation. The trading results should therefore not be interpreted as evidence of universally exploitable or deployable trading profitability.

More broadly, the study illustrates how explainability can contribute to financial machine learning by providing a structured representation of model attribution and by motivating subsequent evaluation of the predictive implications of that attribution. In this sense, feature family analysis complements individual feature explanations by shifting part of the interpretive focus toward how predictive relevance is distributed across related groups of financial variables.

7 Limitations and Future Research

7.1 Study Limitations

Several limitations should be considered when interpreting the findings of the present study. First, the empirical analysis is conducted using a screened modeling universe of equities from the TSE. Consequently, the observed feature family importance patterns are conditional on this selected universe and should not be generalized to the entire TSE or to other financial markets without additional evidence. The clustering analysis provides little evidence that the observed feature family importance profiles are systematically aligned with conventional industry classifications, but this absence of industry alignment does not establish that the identified structure represents a market wide phenomenon.

Second, the proposed framework is constructed using engineered variables derived from daily OHLCV data. Macroeconomic variables, firm fundamentals, textual sentiment, news based information, and alternative data sources are not included in the analysis. Accordingly, the reported feature family structure describes the organization of predictive relevance within the selected feature environment rather than across all potentially relevant sources of financial information.

Third, the feature family taxonomy represents a study specific operational organization of the engineered feature space. Although mutually exclusive family assignments

facilitate aggregation and interpretation, some individual variables may plausibly relate to more than one financial dimension. The resulting family level attribution patterns are therefore conditional on the feature definitions and family assignments adopted in the present study.

Fourth, the empirical design evaluates a fixed prediction horizon within a single historical sample period. The time safe evaluation protocol preserves the chronological roles of training, validation, and internal test observations, but the analysis does not explicitly examine regime dependent behavior, concept drift, rolling feature family importance, or the temporal evolution of the identified attribution patterns. Financial relationships may change across market conditions, and the reported results should therefore not be interpreted as permanently stable over time.

Fifth, the paired statistical inference is conducted across equity level observations that may retain cross sectional dependence arising from common market conditions. Although the Wilcoxon signed rank tests and false discovery rate adjustment provide a systematic assessment of paired performance differences, the resulting significance tests should not be interpreted as inference based on strictly independent securities. This dependence represents a limitation of the cross sectional statistical analysis.

The trading based validation is also subject to implementation limitations. Although transaction costs are incorporated and execution follows the predefined open to open convention, the analysis does not model market impact, order book constraints, or live execution uncertainty. In addition, the trading comparisons are descriptive and are not intended to establish statistically significant or deployable trading superiority.

Finally, the study identifies predictive associations rather than causal mechanisms. Explainability methods, feature family aggregation, and the subsequent predictive evaluations characterize how model attribution and predictive performance are related within the examined feature space, but they do not establish why these relationships arise or whether the identified feature families directly cause subsequent market movements.

Taken together, these limitations define the scope within which the reported findings should be interpreted and identify several directions for further investigation.

7.2 Future Research

The proposed framework opens several directions for future research on interpretable financial machine learning and predictive information organization.

A first priority is external replication across alternative financial environments. Applying the framework to developed equity markets, frontier markets, cryptocurrencies, commodity markets, and other asset classes would allow researchers to examine whether similar feature family importance patterns emerge across different institutional and market settings. Comparative analyses could help distinguish findings that generalize across markets from those that are specific to particular data environments, market structures, or modeling universes.

A second direction concerns the temporal stability of feature family importance and its relationship with prediction horizon. Future studies could investigate rolling

window explainability, dynamic feature family attribution, multiple prediction horizons, and regime dependent predictive structures under changing market conditions. Such analyses would clarify whether dominant feature families remain stable over time or vary across different market regimes and forecasting horizons.

A third direction involves extending both the feature environment and the feature family taxonomy. Incorporating macroeconomic variables, firm fundamentals, textual sentiment, news based information, and alternative data sources would allow predictive information to be studied across a broader range of financial information sources. Future studies could also examine alternative family definitions, overlapping feature groupings, or data driven grouping procedures to evaluate the sensitivity of family level conclusions to the taxonomy used for aggregation.

Finally, future research may investigate the causal and theoretical foundations of the observed predictive patterns. Combining explainable machine learning with causal inference, structural economic modeling, or higher frequency market data may help distinguish predictive associations from underlying economic mechanisms and clarify why particular dimensions of financial information acquire predictive relevance.

Overall, extending the proposed framework across markets, data environments, prediction horizons, temporal regimes, and alternative feature organizations may provide a broader understanding of how model attribution and predictive performance are organized across financial feature spaces.

8 Conclusion

This study investigated the organization of predictive information within a screened universe of Tehran Stock Exchange equities using a time safe interpretable machine learning framework. Rather than treating equity prediction solely as a forecasting task, the study examined whether predictive relevance is uniformly distributed across financial features or systematically concentrated within interpretable feature families.

The empirical evidence reveals a clear concentration of family level attribution within the examined feature space. Daily Price Structure receives the largest aggregate attribution share and is identified as the dominant family for the large majority of evaluated equities, followed by momentum and volume flow and liquidity. The relative ordering of all five feature families is identical under permutation importance and SHAP attribution, indicating that the family ranking is consistent across the two attribution methods examined. The prominence of Daily Price Structure also persists after adjusting family attribution for differences in the number of represented features across families, indicating that this pattern is not explained by unequal family sizes.

The ablation analysis provides an important qualification to this attribution structure. At the aggregate level, the DOM and ALL configurations exhibit similar predictive performance, whereas excluding the dominant family under the NONDOM configuration is associated with substantially lower performance. However, this pattern is not uniform across dominant feature families. Daily Price Structure dominated equities exhibit strong standalone performance under the DOM configuration, while the momentum dominated subgroup shows substantially weaker standalone performance. These findings demonstrate that attributional dominance and standalone predictive

performance are related but distinct properties: the family receiving the greatest model attribution is not necessarily sufficient in isolation to reproduce the predictive performance of the broader feature space.

Predictive statistical inference supports the observed performance differences while also emphasizing their magnitude. DOM differs significantly from NONDOM and ALL differs significantly from NONDOM for both ROC AUC and PR AUC after false discovery rate adjustment. Although DOM also differs significantly from ALL for ROC AUC, the magnitude of this difference is small, while the corresponding PR AUC difference is not statistically significant. The subsequent trading based validation provides complementary descriptive evidence, with the DOM configuration exhibiting the highest median total return and Sharpe ratio both across the evaluated trading sample and within the Daily Price Structure dominated subgroup. These trading results are interpreted as descriptive validation rather than evidence of statistically significant or deployable trading superiority.

The clustering analysis addresses a separate question concerning industry alignment. The weak correspondence between the resulting clusters and conventional industry classifications provides little evidence that feature family importance profiles are systematically partitioned according to industry membership. This finding does not establish a market wide predictive structure, but indicates that conventional industry classifications do not provide a strong explanation for the grouping observed in the feature family importance space.

The findings should be interpreted within the scope of the empirical design. They are conditional on the screened modeling universe, daily OHLCV derived feature space, study specific feature family taxonomy, selected prediction horizon, historical sample, and modeling and evaluation procedures employed in the analysis. The study identifies predictive associations rather than causal mechanisms, and the cross sectional statistical analysis may retain dependence across equities. Further research across markets, prediction horizons, temporal regimes, information sources, and alternative feature family definitions is therefore required to assess the broader generalizability of the observed patterns.

Overall, the principal contribution of the study is a family level empirical framework that moves beyond isolated feature attribution by jointly examining how model attribution is organized across interpretable feature families and what attributional dominance implies for subsequent predictive performance. Within the examined setting, Daily Price Structure emerges as the most prominent attribution family, but the ablation results demonstrate that attributional dominance does not imply uniform standalone predictive performance across dominant families. More broadly, the study shows how explainability, when combined with feature family aggregation and predictive evaluation, can provide a structured perspective on the organization and predictive implications of model attribution in financial machine learning.

A Feature Construction and Indicator Definitions

This appendix reports the decision criteria used in the two stage screening procedure and documents the feature construction rules employed in the empirical framework.

The first screening table presents the data quality, liquidity related, and preliminary predictive feasibility criteria applied to the adjusted OHLCV data. The second screening table reports the machine learning feasibility criteria used to determine the final modeling universe.

A.1 Data Quality and Predictive Feasibility Thresholds

The empirical universe was constructed through two sequential screening stages. Only equities classified as PASS in the first stage quality control gate were evaluated in the second stage machine learning feasibility gate. Similarly, only equities classified as PASS in the second stage entered the final modeling universe.

Table 16 reports the criteria and thresholds applied in the first stage of the screening procedure.

Table 16: First Stage Data Quality and Preliminary Feasibility Thresholds

Criterion	Threshold	Screening Role
Historical observations	$N \geq 700$	Minimum estimation history
Missing observations in core OHLCV fields	$\leq 2\%$	Raw-data completeness
Zero-volume observations	$\leq 5\%$	Liquidity requirement
Locked-range observations ($H_t = L_t$)	$\leq 8\%$	Detection of prolonged price stagnation
Positive-class proportion	$0.35 \leq p_1 \leq 0.65$	Target-class balance
Mean time-series cross-validated ROC-AUC	≥ 0.55	Preliminary predictive feasibility
ROC-AUC improvement over the best baseline	≥ 0.02	Incremental predictive content
Standard deviation of cross-validated ROC-AUC	≤ 0.08	Temporal stability
Valid validation folds	At least 3 of 5	Reliability of cross-validation estimates

Table 17 reports the corresponding criteria and thresholds applied in the second stage machine learning feasibility gate.

The remainder of this appendix documents the definitions and mathematical formulations of the engineered features used in the empirical analysis. All variables were constructed from daily adjusted OHLCV data. Let O_t , H_t , L_t , C_t , F_t , and V_t denote adjusted open, high, low, close, final adjusted price, and trading volume at date t , respectively. All lagged, rolling, and ratio based variables were computed using only information available up to date t . The assignment of the engineered features to the five feature families used in the empirical analysis is documented separately in Appendix B.

Table 17: Second Stage Machine Learning Feasibility Thresholds

Criterion	Threshold	Purpose
Usable observations after feature construction	$N_{ML} \geq 500$	Sufficient estimation sample
Positive-class proportion	$0.35 \leq p_1 \leq 0.65$	Target-class balance
Mean time-series cross-validated ROC-AUC	≥ 0.55	Discriminatory performance
Mean time-series cross-validated average precision	≥ 0.55	Precision-recall performance
Standard deviation of ROC-AUC	≤ 0.10	Temporal stability
Valid validation folds	At least 3 of 5	Reliability of cross-validation estimates
Standardized-feature clipping	$[-3, 3]$	Reduction of extreme-value influence

Notes: Both stages used five-fold expanding-window `TimeSeriesSplit` validation and balanced Logistic Regression. Equities classified as BORDERLINE or FAIL at either applicable stage were excluded from the subsequent modeling universe.

A.2 Safe Division Rule

For ratio based variables, division was implemented using a safe division rule:

$$\text{safe_divide}(a, b) = \begin{cases} a/b, & b \neq 0 \text{ and } b \text{ is finite,} \\ \text{NaN}, & \text{otherwise.} \end{cases}$$

Unless otherwise stated, ratio based expressions reported below were implemented using this safe division rule.

A.3 Price Derived Variables

$$\text{Price_Range}_t = H_t - L_t$$

$$\text{Body_Range}_t = |C_t - O_t|$$

$$\text{Final_Range}_t = |C_t - F_t|$$

$$\text{Upper_Shadow}_t = H_t - \max(C_t, O_t)$$

$$\text{Lower_Shadow}_t = \min(C_t, O_t) - L_t$$

$$\text{CLV}_t = \frac{(C_t - L_t) - (H_t - C_t)}{H_t - L_t}$$

A.4 Volume and Liquidity Variables

For $k \in \{1, 2, 3\}$,

$$\text{Vol_Change}_{k,t} = V_t - V_{t-k}.$$

For $w \in \{3, 7\}$, the volume average ratio is defined as

$$\text{Vol_Avg_Ratio}_{w,t} = \frac{V_t}{\frac{1}{w} \sum_{i=0}^{w-1} V_{t-i}}.$$

The typical price is

$$TP_t = \frac{H_t + L_t + F_t}{3}.$$

The seven day rolling volume weighted price is

$$\text{W_Price}_t = \frac{\sum_{i=0}^6 TP_{t-i} V_{t-i}}{\sum_{i=0}^6 V_{t-i}}.$$

On Balance Volume is initialized at zero and updated as

$$\text{OBV}_t = \begin{cases} \text{OBV}_{t-1} + V_t, & F_t > F_{t-1}, \\ \text{OBV}_{t-1} - V_t, & F_t < F_{t-1}, \\ \text{OBV}_{t-1}, & F_t = F_{t-1}. \end{cases}$$

For the Money Flow Index, money flow is defined as

$$MF_t = TP_t V_t.$$

Positive and negative money flows are

$$MF_t^+ = MF_t \cdot \mathbf{1}(TP_t > TP_{t-1})$$

and

$$MF_t^- = MF_t \cdot \mathbf{1}(TP_t \leq TP_{t-1}).$$

The seven day money flow ratio is

$$MFR_t = \frac{\sum_{i=0}^6 MF_{t-i}^+}{\sum_{i=0}^6 MF_{t-i}^-},$$

and the Money Flow Index is

$$\text{MFI}_t = 100 - \frac{100}{1 + MFR_t}.$$

Chaikin Money Flow is defined as

$$\text{CMF}_t = \frac{\sum_{i=0}^6 \text{CLV}_{t-i} V_{t-i}}{\sum_{i=0}^6 V_{t-i}}.$$

A.5 Lagged Return and Momentum Variables

For $k \in \{1, 2, 3, 5, 7\}$,

$$\text{Final_Lag}_{k,t} = F_{t-k}.$$

Percentage price changes over the corresponding horizons are defined as

$$\text{Pct_Change}_{k,t} = \frac{F_t - F_{t-k}}{F_{t-k}}, \quad k \in \{1, 2, 3, 5, 7\}.$$

The ten day momentum variable is

$$\text{Momentum_10}_t = \frac{F_t}{F_{t-10}} - 1.$$

The seven day Rate of Change is

$$\text{ROC}_t = 100 \times \frac{F_t - F_{t-7}}{F_{t-7}}.$$

A.6 Rolling, Volatility, and Standardized Variables

$$\text{Rolling_Mean_5}_t = \frac{1}{5} \sum_{i=0}^4 F_{t-i}$$

$$\text{Rolling_Std_10}_t = \text{sd}(F_t, F_{t-1}, \dots, F_{t-9})$$

$$\bar{F}_{10,t} = \frac{1}{10} \sum_{i=0}^9 F_{t-i}$$

$$\text{Z_Score_10}_t = \frac{F_t - \bar{F}_{10,t}}{\text{Rolling_Std_10}_t}$$

$$\text{Volatility_Break}_t = \frac{H_t - L_t}{C_{t-1}}$$

The true range is defined as

$$TR_t = \max \{H_t - L_t, |H_t - F_{t-1}|, |L_t - F_{t-1}|\}.$$

The seven day Average True Range is

$$\text{ATR}_t = \frac{1}{7} \sum_{i=0}^6 TR_{t-i}.$$

A.7 Trend Variables

$$\text{Trend.Up.5}_t = \frac{1}{5} \sum_{i=0}^4 \mathbf{1}(F_{t-i} - F_{t-i-1} > 0)$$

$$\text{SMA.7}_t = \frac{1}{7} \sum_{i=0}^6 F_{t-i}$$

The MACD variable uses exponential moving averages with spans 3 and 7:

$$\text{MACD}_t = \text{EMA}_{3,t}(F) - \text{EMA}_{7,t}(F).$$

The corresponding signal variable is

$$\text{MACD.Signal}_t = \text{EMA}_{7,t}(\text{MACD}).$$

A.8 Relative Strength Index

The daily change in final adjusted price is

$$\Delta F_t = F_t - F_{t-1}.$$

Positive and negative price movements are defined as

$$G_t = \max(\Delta F_t, 0), \quad L_t^{RSI} = \max(-\Delta F_t, 0).$$

Using seven day rolling averages,

$$RS_t = \frac{\overline{G}_{7,t}}{\overline{L}_{7,t}^{RSI}},$$

and the Relative Strength Index is

$$\text{RSI}_t = 100 - \frac{100}{1 + RS_t}.$$

A.9 Feature Construction Policy

All variables were computed chronologically. Lagged and rolling transformations used only past and current observations. Ratio based variables were evaluated using the safe division rule defined above. For RSI and MFI, invalid or infinite values were replaced with the neutral value 50, consistent with the implementation. The resulting engineered feature set was subsequently assigned to the predefined feature families documented in Appendix B.

B Feature Family Taxonomy

This appendix documents the feature family taxonomy used throughout the empirical analysis. Feature level importance values obtained from the explainability methods were aggregated into broader feature families using a deterministic token based assignment procedure.

Each of the 37 engineered features was assigned to a single feature family according to predefined token matching rules. The taxonomy was specified before the explainability analysis and remained fixed throughout the study. All features were assigned to one of five mutually exclusive families, and no feature remained unassigned in the final taxonomy.

B.1 Momentum Family

The Momentum family captures directional persistence, recent price changes, and return based behavior. Table 18 lists the features assigned to this family.

Table 18: Momentum Family Features

Feature	Description
Final_Lag_1	One day lagged final adjusted price
Final_Lag_2	Two day lagged final adjusted price
Final_Lag_3	Three day lagged final adjusted price
Final_Lag_5	Five day lagged final adjusted price
Final_Lag_7	Seven day lagged final adjusted price
Pct_Change_1	One day percentage price change
Pct_Change_2	Two day percentage price change
Pct_Change_3	Three day percentage price change
Pct_Change_5	Five day percentage price change
Pct_Change_7	Seven day percentage price change
Momentum_10	Ten day price momentum
ROC	Seven day Rate of Change
RSI	Seven day Relative Strength Index

B.2 Volatility Family

The Volatility family captures price dispersion, variability, normalized deviations, and daily range behavior. Table 19 lists the features assigned to this family.

Table 19: Volatility Family Features

Feature	Description
ATR	Seven day Average True Range
Rolling_Std_10	Ten day rolling standard deviation
Z_Score_10	Standardized deviation from the ten day rolling mean
Price_Range	Daily high minus low price range
Volatility_Break	Daily price range normalized by the previous closing price

B.3 Volume Flow and Liquidity Family

The Volume Flow and Liquidity family captures trading activity, volume dynamics, weighted price behavior, and money flow conditions. Table 20 lists the features assigned to this family.

Table 20: Volume Flow and Liquidity Features

Feature	Description
Vol_Change_1	One day change in trading volume
Vol_Change_2	Two day change in trading volume
Vol_Change_3	Three day change in trading volume
Vol_Avg_Ratio_3	Volume relative to its three day rolling average
Vol_Avg_Ratio_7	Volume relative to its seven day rolling average
OBV	On Balance Volume
CMF	Chaikin Money Flow
MFI	Money Flow Index
W_Price	Volume weighted price measure

B.4 Trend Family

The Trend family captures smoothed price behavior, directional consistency, and lower frequency price structure. Table 21 lists the features assigned to this family.

Table 21: Trend Features

Feature	Description
Rolling_Mean_5	Five day rolling mean of the final adjusted price
SMA_7	Seven day simple moving average
MACD	Difference between short and long exponential moving averages
MACD_Signal	Exponential moving average of MACD
Trend_Up_5	Fraction of positive daily price movements over five days

B.5 Daily Price Structure Family

The Daily Price Structure family captures localized candlestick structure and short horizon price configuration derived from daily OHLCV observations. Table 22 lists the features assigned to this family.

Table 22: Daily Price Structure Family Features

Feature	Description
Body_Range	Absolute candlestick body magnitude
Final_Range	Absolute difference between closing and final adjusted prices
Upper_Shadow	Upper candlestick shadow length
Lower_Shadow	Lower candlestick shadow length
CLV	Closing location within the daily high–low range

References

- Aas, K., Jullum, M., Løland, A. (2021). Explaining individual predictions when features are dependent: More accurate approximations to shapley values. *Artificial Intelligence*, 298, 103502, <https://doi.org/10.1016/j.artint.2021.103502>
- Achelis, S.B. (2001). *Technical analysis from a to z*. McGraw-Hill.
- Bekaert, G., & Harvey, C.R. (2003). Emerging markets finance. *Journal of Empirical Finance*, 10(1–2), 3–55, [https://doi.org/10.1016/S0927-5398\(02\)00054-3](https://doi.org/10.1016/S0927-5398(02)00054-3)
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327, [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1)
- Box, G.E.P., Jenkins, G.M., Reinsel, G.C., Ljung, G.M. (2015). *Time series analysis: Forecasting and control* (5th ed.). Hoboken, NJ: John Wiley & Sons.
- Černevičienė, J., & Kabašinskas, A. (2024). Explainable artificial intelligence (XAI) in finance: A systematic literature review. *Artificial Intelligence Review*, 57, 216, <https://doi.org/10.1007/s10462-024-10854-8>
- Fama, E.F., & French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56, [https://doi.org/10.1016/0304-405X\(93\)90023-5](https://doi.org/10.1016/0304-405X(93)90023-5)
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669, <https://doi.org/10.1016/j.ejor.2017.11.054>

- Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273, <https://doi.org/10.1093/rfs/hhaa009>
- Harvey, C.R. (1995). Predictable risk and returns in emerging markets. *Review of Financial Studies*, 8(3), 773–816, <https://doi.org/10.1093/rfs/8.3.773>
- Hoepner, A.G.F., McMillan, D., Vivian, A., Simen, C.W. (2021). Significance, relevance and explainability in the machine learning age: An econometrics and financial data science perspective. *The European Journal of Finance*, 27(1–2), 1–7, <https://doi.org/10.1080/1351847X.2020.1847725>
- Lee, K.H., & Jo, G.S. (1999). Expert system for predicting stock market timing using a candlestick chart. *Expert Systems with Applications*, 16(4), 357–364, [https://doi.org/10.1016/S0957-4174\(99\)00011-1](https://doi.org/10.1016/S0957-4174(99)00011-1)
- Lundberg, S.M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems* (Vol. 30).
- Murphy, J.J. (1999). *Technical analysis of the financial markets*. New York Institute of Finance.
- Owen, A.B., & Prieur, C. (2017). On shapley value for measuring importance of dependent inputs. *SIAM/ASA Journal on Uncertainty Quantification*, 5(1), 986–1002, <https://doi.org/10.1137/16M1097717>
- Ribeiro, M.T., Singh, S., Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 1135–1144). ACM.
- Sezer, O.B., Gudelek, M.U., Ozbayoglu, A.M. (2020). Financial time series forecasting with deep learning: A systematic literature review. *Applied Soft Computing*, 90, 106181, <https://doi.org/10.1016/j.asoc.2020.106181>
- Sirignano, J., & Cont, R. (2019). Universal features of price formation in financial markets: Perspectives from deep learning. *Quantitative Finance*, 19(9), 1449–1459, <https://doi.org/10.1080/14697688.2019.1622293>
- Strobl, C., Boulesteix, A.-L., Zeileis, A., Hothorn, T. (2007). Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC*

Weber, P., Carl, K.V., Hinz, O. (2024). Applications of explainable artificial intelligence in finance: A systematic review of finance, information systems, and computer science literature. *Management Review Quarterly*, 74, 867–907, <https://doi.org/10.1007/s11301-023-00320-0>

Yeo, W.J., van der Heever, W., Mao, R., Cambria, E., Satapathy, R., Mengaldo, G. (2025). A comprehensive review on financial explainable ai. *Artificial Intelligence Review*, 58, 189, <https://doi.org/10.1007/s10462-024-11077-7>

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

Both authors contributed equally to the conception and design of the study, methodology development, data analysis, interpretation of the results, and manuscript preparation. Both authors reviewed, revised, and approved the final version of the manuscript.

Data Availability

The raw market data analyzed in this study were retrieved using the publicly available `finpy-tse` Python library. The processed datasets generated during the study are available from the corresponding author upon reasonable request. Supporting empirical outputs underlying the reported analyses are provided in Online Resource 1.

Code Availability

The analysis notebook documenting the computational workflow is provided in Online Resource 1. The Python source code developed for this study will be made publicly available upon publication. Additional code materials are available from the corresponding author upon reasonable request.

Use of Artificial Intelligence Tools

During manuscript preparation, ChatGPT (OpenAI) was used to assist with language refinement, wording, and editorial revision of author prepared text. The authors reviewed and revised all AI-assisted output and take full responsibility for the content of the final manuscript. The tool was not used to generate the empirical results reported in the study.

Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.